KARAN KUMAR SOUNDRAPANDIAN

+91 9600636429 thisiskaree@gmail.com LinkedIn GitHub Chennai, Tamil Nadu

Education

Thapar Institute of Engineering and Technology, Patiala **2024 – 2028**
B.Tech, Electronics and Computer Engineering *Expected Graduation: 2028*

Kendriya Vidyalaya Ashok Nagar, Chennai
Senior Secondary Education (Class XII), CBSE

Projects

Project Obscura MERN Stack
- Developed a community-driven photography platform using **MongoDB, Express.js, React.js, and Node.js** for creative challenges, image submissions, authentication, and creator engagement.

Route Optimization Visualizer Python, Algorithms
- Built an interactive graph-optimization visualizer implementing **Dijkstra's Algorithm, A*, TSP, and MST** with dynamic city, distance, and path visualization.

Network Packet Analyzer Python, Networking
- Developed a Wireshark-style packet analyzer for inspecting network traffic, identifying protocols, and extracting IP addresses, ports, and packet-level information.

Neural Network From Scratch Python, NumPy, Machine Learning
- Engineered a neural-network framework with **Linear, ReLU, Sigmoid, and Softmax** layers, loss functions, gradient checking, and SGD, Momentum, and Adam optimizers.

Production-Grade CI/CD Cloud Deployment Platform Docker, Kubernetes, GitHub Actions
- Engineered automated CI/CD workflows with testing, Docker containerization, Kubernetes deployment, health checks, rolling updates, autoscaling, rollback, and security scanning.

Customer Churn Prediction Python, Scikit-learn, XGBoost, SHAP
- Built an end-to-end churn prediction pipeline using **Logistic Regression, Random Forest, XGBoost, feature engineering, and SHAP** for interpretable customer-risk analysis.

Advanced Retail Customer Intelligence Platform SQL, Python, Power BI
- Developed a retail analytics platform using **SQL, Python, Pandas, NumPy, and Power BI** for customer segmentation, RFM analysis, revenue trends, product performance, and retention insights.

EV Two-Wheeler Market Entry Strategy Python, Excel, Power BI
- Created a data-driven EV market-entry strategy using competitive analysis, customer segmentation, market research, quantitative analysis, and Power BI-based insights.

Expense Tracker MERN Stack
- Built a full-stack expense management application with **JWT authentication, role-based access, budgeting, analytics, search, filtering, Excel export, REST APIs**, and Vercel/Render deployment.

Geometrical Shape Detection and Recognition Python, OpenCV
- Developed a computer-vision application using **OpenCV, contour detection, edge detection, image analysis, and object recognition** to identify and classify geometric shapes.

Technical Skills

Languages: Python, JavaScript, C++, C, SQL, HTML5, CSS3
Web Development: React.js, Node.js, Express.js, REST APIs, JWT, MVC, CRUD
Databases: MongoDB, MySQL, Database Design, Data Modeling, Query Optimization
ML / AI: NumPy, Pandas, Scikit-learn, XGBoost, Random Forest, Logistic Regression, SHAP, Neural Networks, Generative AI, LLMs
Computer Vision: OpenCV, Shape Detection, Image Recognition, Contour Analysis, Edge Detection, Object Classification
DevOps / Cloud: GitHub Actions, Docker, Kubernetes, CI/CD, AWS, Vercel, Render, Prometheus, Grafana, Trivy
Tools: Git, GitHub, Postman, VS Code, npm, Chrome Developer Tools
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems, Software Engineering
Electronics: PCB Design and Fabrication, Circuit Analysis, Analog Electronics, Digital Electronics, Embedded Systems

Certifications

- Top 25 Nominee – IFP 50-Hour Filmmaking Challenge (2025)
- Python for Data Science, AI & Development – IBM
- Introduction to Computer Networking – Coursera
- Introduction to Large Language Models – Google Cloud Skills Boost
- Geometrical Shape Detection and Recognition Using Python in Image Processing
- PCB Design and Fabrication

Leadership & Achievements

- **General Secretary, Fine Arts and Photography Society (FAPS), TIET** – 2026–27; lead creative initiatives, photography activities, and society events.
- Selected among the **Top 25 Nominees** in the IFP 50-Hour Filmmaking Challenge; actively involved in photography, filmmaking, and creative technology projects.